[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

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Class 3 Maths — CA-1 (Hard) — Answer Key

Class : 3
Subject : Maths (Math-Mela)

Max Marks : 30
Duration : 1 Hour

Syllabus: Chapters 1–3 — What's in a Name?, Toy Joy, Double Century

General Instructions: All questions are compulsory. For Section A, write the correct option letter.

ANSWER KEY

SECTION A — Multiple Choice Questions (1 Mark each)

1. A basket has 187 mangoes. If they are packed into boxes of 10, how many full boxes can be made and how many mangoes are left over? *(1 Mark)*

A. 17 boxes, 7 left over
B. 18 boxes, 7 left over ✓
C. 18 boxes, 0 left over
D. 19 boxes, 7 left over

2. Which shape, when cut straight through the middle parallel to its base, always shows a circle as the cut surface? *(1 Mark)*

A. Cube
B. Cuboid
C. Cylinder ✓
D. None of these

3. A cuboid has length, breadth, and height all different from each other. How many pairs of identical (equal) faces does it have? *(1 Mark)*

A. 1
B. 2
C. 3 ✓
D. 6

4. If the number 1_5 (tens digit missing) is greater than 143 but less than 168, which digit could go in the blank? *(1 Mark)*

A. 3
B. 5 ✓
C. 2
D. 9

5. What is the greatest 3-digit number you can make that is less than 200 using the digits 1, 9, and 8 (using each digit exactly once)? *(1 Mark)*

A. 198 ✓
B. 189
C. 918
D. 981

6. A toy is made by joining a cone on top of a cylinder. How many total flat faces does the combined toy have (counting only the outer visible flat faces)? *(1 Mark)*

A. 1
B. 2 ✓
C. 3
D. 4

7. Which of these statements is TRUE about all cubes? *(1 Mark)*

A. A cube is never also a cuboid
B. Every cube is a special type of cuboid ✓
C. A cube has curved surfaces
D. A cube has only 4 faces

SECTION B — Short Answer Questions (2 Marks each)

1. A pond has 176 fish. If 38 fish are caught and 22 new fish are added, how many fish are in the pond now? *(0 Marks)*

➤ 176 - 38 = 138, then 138 + 22 = 160 fish are in the pond now.

2. Explain why a sphere cannot be stacked like a cube, using the idea of flat and curved surfaces. *(0 Marks)*

➤ A sphere has only a curved surface and no flat faces at all, so there is no flat part for another object to rest on without rolling away, unlike a cube which has flat faces that let objects sit steadily on top of each other.

3. Write a 3-digit number where the hundreds digit is double the tens digit, and the ones digit is 5. Explain your reasoning. *(0 Marks)*

➤ If the tens digit is 1, the hundreds digit must be double that, which is 2; with the ones digit given as 5, the number is 215. Check: hundreds digit 2 is double the tens digit 1, and the ones digit is 5, so 215 fits all the conditions.

4. Two toy blocks, a cube and a cuboid, look similar from one side. Explain one way you could tell them apart just by looking at all their faces. *(0 Marks)*

➤ You can check if all six faces are exactly equal squares - if yes, it is a cube; if the faces are rectangles of different sizes (length longer than breadth or height), it is a cuboid, not a cube.

5. A number lies between 150 and 200. Its tens digit is 7 and its ones digit is the same as its tens digit. What is the number? *(0 Marks)*

➤ The number is 177 (hundreds digit 1, tens digit 7, ones digit 7, and 177 lies between 150 and 200).

6. A rectangular toy box (cuboid) is placed so its largest face touches the ground. Explain why this makes the box more stable than standing it on its smallest face. *(0 Marks)*

➤ A larger flat face touching the ground gives the box a wider base to balance on, making it less likely to tip over, while a smaller face gives a narrower, less stable base.

SECTION C — Short Answer Questions (Explain) (3 Marks each)

1. A trader has 200 oranges. Every day, the trader sells some oranges and this pattern continues: Day 1 sells 34, Day 2 sells 12 more than Day 1, and Day 3 sells 8 fewer than Day 2. Find the total oranges sold over the three days and the oranges left after Day 3. *(0 Marks)*

➤ Day 1: 34 oranges sold.

➤ Day 2: 34 + 12 = 46 oranges sold.

➤ Day 3: 46 - 8 = 38 oranges sold.

➤ Total sold = 34 + 46 + 38 = 118 oranges.

➤ Oranges left = 200 - 118 = 82 oranges.

2. Explain, using properties of faces, edges and corners, how you would prove to a younger student that a cube and a cuboid are related but not exactly the same shape. *(0 Marks)*

➤ Both shapes have 6 faces, 12 edges, and 8 corners, which makes them look similar in structure.

➤ However, in a cube every face is an identical square, while in a cuboid, opposite faces are equal rectangles but not necessarily squares, and not all faces need to be the same size.

➤ You could show this by comparing a dice (cube) to a brick or book (cuboid) side by side, pointing out that the dice looks the same from every direction, but the brick looks longer from one side.

SECTION D — Long Answer Questions (5 Marks each)

1. A large box contains 200 building blocks made up of cubes, cuboids, cylinders, cones, and spheres. If there are 40 cubes, 35 cuboids, 45 cylinders, and 30 cones, find how many spheres are in the box, and then explain one property that makes spheres different from all the other four shapes in this box. *(0 Marks)*

➤ Total non-sphere blocks = 40 + 35 + 45 + 30 = 150.

➤ Number of spheres = 200 - 150 = 50 spheres.

➤ A sphere is different from the other four shapes because it has absolutely no flat faces or edges - the entire surface is curved, while cubes, cuboids, cylinders, and cones all have at least one flat face.

➤ This means a sphere can roll in every direction, while the others either cannot roll (cube, cuboid) or roll only in a limited way (cylinder, cone).